

From Gaming Commodity to AI Hard Currency: GPU Pricing Dynamics and Consumption Pattern Shifts in the Pre-LLM and LLM Eras

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Abstract

The emergence of Large Language Models (LLMs) has fundamentally transformed the GPU market, shifting its primary demand driver from gaming enthusiasts to AI infrastructure providers. This paper examines the dramatic price evolution and consumption pattern restructuring across the pre-LLM and LLM eras. We document how flagship GPU pricing escalated from approximately \$500 in the GTX 1080 era to over \$1,500 for current-generation models, while older compute GPUs such as the A100 have appreciated more than 150% above their original launch prices, defying traditional consumer electronics depreciation patterns. Through analysis of market data, semiconductor supply chain constraints, and shifting corporate capital expenditure allocations, we identify three structural drivers of this transformation: the exponential growth in AI compute demand, HBM memory supply bottlenecks, and the financialization of GPUs as collateralizable assets. Furthermore, we analyze the bifurcated consumer landscape where enterprise buyers exhibit near-zero price sensitivity while gaming consumers face forced downgrading. Finally, we discuss the emerging shift from training to inference workloads and its implications for market structure, including the rising market share of Chinese domestic GPU alternatives. Our findings suggest that the GPU market has transitioned from a Moore’s Law-driven gradual improvement regime to a supply-constrained scarcity economy, permanently altering the relationship between hardware performance and market valuation.

1 Introduction

The graphics processing unit (GPU) has undergone a remarkable identity transformation over the past decade. Once primarily marketed as an enthusiast component for PC gaming, it has become the foundational hardware underpinning the artificial intelligence revolution. The release of ChatGPT in late 2022 marked a watershed moment,

catalyzing an unprecedented surge in demand for computational infrastructure that has fundamentally restructured the economics of the semiconductor industry (Huang, 2024).

This paper investigates the price dynamics and consumption pattern shifts that distinguish the pre-LLM era from the LLM era. We argue that three fundamental changes have occurred. First, the demand base has expanded from approximately 200 million PC gamers to virtually every enterprise engaged in AI research and deployment, creating what industry observers have termed a “GPU arms race” (Gelonghui, 2025). Second, the supply side has become increasingly constrained, with HBM memory production capacity becoming a critical bottleneck that affects the entire GPU ecosystem (TrendForce, 2025). Third, the GPU has transitioned from a depreciating consumer good to an appreciating financial asset, with leasing rates and resale values defying traditional electronics depreciation curves (SemiAnalysis, 2026).

The paper is organized as follows. Section 2 reviews the pre-LLM era, establishing the baseline pricing and consumption patterns. Section 3 details the LLM-era transformation, examining price escalation, corporate procurement behavior, and the bifurcation of consumer markets. Section 4 presents empirical data through figures and tables. Section 5 analyzes emerging trends, including the shift toward inference workloads and the rise of domestic alternatives. Section 6 concludes with implications for consumers, enterprises, and policymakers.

2 The Pre-LLM Era: Gaming as the Sole Anchor

2.1 Price Evolution and the “Golden Ratio”

During the decade preceding the LLM revolution, GPU pricing followed a relatively predictable trajectory governed by Moore’s Law and its corollary in graphics performance. The industry operated under what we term the “Golden Ratio” pricing framework: the performance per dollar doubled approximately every 18 to 24 months, while flagship product prices remained anchored around the \$500 to \$700 range.

The GeForce GTX 1080, released in 2016, exemplified this paradigm. At a launch price of \$599, it delivered approximately 9 TFLOPS of FP32 compute performance, representing a roughly 70% improvement over its predecessor at a similar price point. The subsequent generation, the RTX 2080 Ti, represented the first significant deviation, launching at \$999 in 2018, which NVIDIA justified through the introduction of ray tracing and tensor core hardware (TechWeb, 2025). However, even this pricing remained within the historical pattern of incremental increases in exchange for substantial new hardware capabilities.

2.2 Consumer Demographics and Market Structure

The pre-LLM consumer landscape was characterized by relative homogeneity. Gaming enthusiasts constituted the primary market segment, with secondary segments including content creators, cryptocurrency miners during periodic booms, and academic researchers using GPUs for scientific computing. The Steam hardware survey, widely considered the most comprehensive snapshot of the gaming PC market, showed that the vast majority of users spent between \$200 and \$500 on graphics cards, with sub-\$300 cards comprising the largest share (TechWeb, 2025).

Competition between NVIDIA and AMD was robust, with both companies releasing performance-tiered product stacks. The availability of multiple options at each price point created a market where consumers could reliably upgrade every two to three generations and expect approximately 30-50% performance gains for similar or slightly higher prices. Cryptocurrency mining booms in 2017 and 2021 created temporary supply disruptions and price spikes, but these proved transitory; as noted by industry observers, the *mining-driven* price increases were fundamentally different from the *AI-driven* price increases that followed (Zhongguancun Online, 2026a). The former was a speculative bubble that inevitably corrected; the latter represents a structural shift in underlying demand.

2.3 The Natural Order: Depreciation as the Default

In the pre-LLM era, GPU depreciation followed a simple and predictable pattern. A new card would typically lose 20-30% of its value within the first year, 50% within two years, and would be largely obsolete within five years. The used market was a buyer's market, and the concept of a GPU appreciating in value was so foreign as to be practically nonexistent. This depreciation pattern was so fundamental to the consumer electronics experience that it shaped consumer behavior, encouraging what became known as the "wait for the next generation" or "go for the eventual price drop" mentality (36Kr, 2026).

The predictability of this cycle allowed consumers to make informed purchasing decisions based on their gaming requirements, their display resolution, and their target frame rates. The relationship between hardware capability and software requirement was well understood: game developers targeted the installed base, and graphics settings were designed to scale across the available hardware spectrum. This created a virtuous cycle where improved hardware enabled better games, which in turn drove hardware upgrades.

3 The LLM Era: A Structural Break

3.1 The Demand Shock and Supply Bottlenecks

The release of ChatGPT in November 2022 initiated what we term a *demand shock* of unprecedented magnitude in the semiconductor industry. LLM training and inference require massive parallel computing capacity, and GPUs, particularly NVIDIA's offerings with their CUDA ecosystem, have proven to be the most efficient hardware for these workloads. The result was a sudden and dramatic shift in demand, as every major technology company began building or expanding their AI infrastructure.

This demand shock encountered a supply side that was ill-prepared for such rapid growth. The semiconductor industry, accustomed to the gradual demand growth of the PC era, could not instantly scale production. The bottleneck was not just in GPU manufacturing capacity but also in memory production; as reported by TrendForce, DRAM contract prices surged 80-100% during this period, with HBM (High Bandwidth Memory) production capacity becoming particularly strained (TrendForce, 2025).

The consequences for GPU pricing were immediate and dramatic. The A100 GPU, released in 2020 at a price of approximately \$15,000, was selling for over \$30,000 in secondary markets by 2023. The H100, released in 2023, commanded prices exceeding \$50,000, with availability severely limited (Zhongguancun Online, 2026a). More remarkably, the phenomenon extends to older generations; industry analysis shows that even

the older V100 GPUs, released in 2017, have experienced sustained price appreciation (Hengzhou Chengsi, 2025).

3.2 The Bifurcation of Consumption

The LLM era has created two distinct GPU markets operating simultaneously but governed by fundamentally different economic principles.

3.2.1 Enterprise Market: The “Hard Currency” Phenomenon

For cloud providers and AI companies, GPUs have transitioned from capital equipment to core production assets. Their purchase decisions are governed not by price sensitivity but by the ability to secure supply. This has led to what industry observers term the “hard currency” phenomenon: GPUs are increasingly treated as collateralizable assets, and computing capacity can be leased or traded as a fungible commodity (SemiAnalysis, 2026). CoreWeave and similar platforms have capitalized on this trend by offering GPU compute as a time-based service, with resources consistently selling out as soon as they become available (Pedaily, 2026).

The shift in corporate capital expenditure has been dramatic. The proportion of data center capital spending allocated to storage components, including HBM and related memory, is projected to exceed 40% by 2026 (SemiAnalysis, 2026). This is reflected in Table 2, which shows the reallocation of investment away from traditional IT infrastructure toward AI-specific hardware.

3.2.2 Consumer Market: The Crowding-Out Effect

The enterprise demand surge has had a pronounced “crowding-out” effect on the consumer market. As wafer capacity and packaging resources are diverted to high-margin enterprise products, consumer GPU supply has been constrained. The result is a significant price escalation across all tiers, with the flagship models that once cost \$700 now routinely exceeding \$1,200.

This pricing pressure has forced consumer behavior changes, as shown in Table 1. Sales data reveals that high-end models such as the RTX 5070 Ti and the RX 9070 XT have fallen out of the top sales ranks, replaced by mid-tier alternatives like the RTX 5070 and RTX 5060 (Zhongguancun Online, 2026b). The traditional *upgrade cycle* has been disrupted, with consumers holding onto older cards for longer. More troublingly, the trend toward 8GB VRAM configurations represents a step backward in gaming memory capacity, reflecting cost pressures rather than technological progression (Sohu, 2026).

The impact on the broader PC market has been substantial. IDC projections indicate an average PC price increase of 4-8% in 2026, with AI PC models accounting for an estimated 55% of the market (IDC, 2026). The previously clear relationship between hardware capability and consumer price has been broken: consumers are paying more for less, or at best, similar performance at inflated prices.

3.3 The “Good Wine” Effect

NVIDIA CEO Jensen Huang has described the current GPU pricing phenomenon as the *good wine effect*, where GPUs, like fine wines, appreciate in value with age rather than depreciating (Huang, 2024). This characterization, while colorful, captures a fundamental

economic reality: the GPU market has transitioned from a *Moore’s Law-driven depreciation model* to a *supply-constrained scarcity model*. In the latter, the age of a GPU is not a determinant of its value; its computational capability and its availability determine its worth.

This effect is most apparent in the enterprise market. Figure 3 illustrates how A100 GPUs, despite being several years old and superseded by H100s, continue to trade at prices substantially above their original launch prices. The appreciation phenomenon extends to leasing rates: rental costs for A100 and H100 instances have risen for five and seven consecutive months respectively as of mid-2026 (Du, 2026).

4 Empirical Evidence

4.1 Price Evolution Figures

We present three figures depicting the GPU price evolution across the pre-LLM and LLM eras. Figure 1 shows the dramatic price escalation of flagship GPUs, with the pre-LLM era maintaining relatively stable pricing around the \$600-\$800 range, followed by a sharp post-2022 increase. Figure 2 illustrates the widening gap between price increases and performance improvements, demonstrating how consumers are increasingly paying more per unit of performance. Figure 3 depicts the secondary market pricing for enterprise GPUs, showing how A100 and H100 have appreciated substantially above their launch prices.

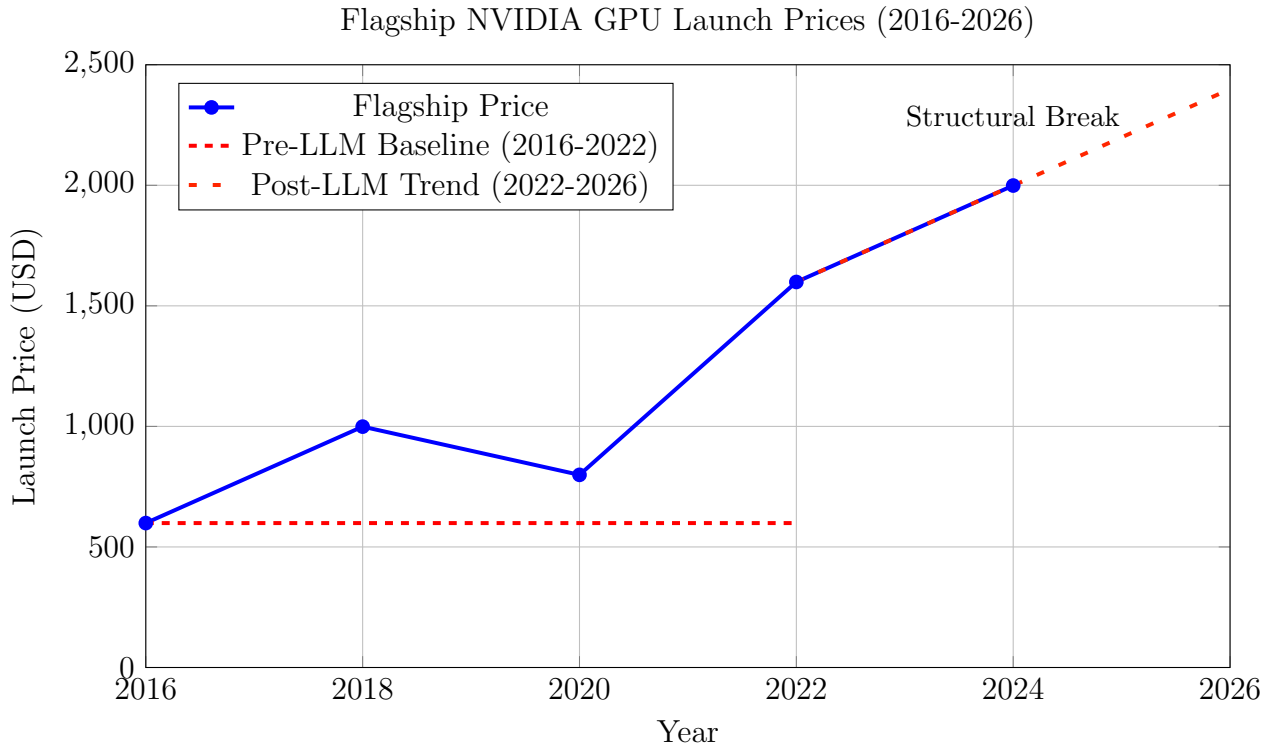


Figure 1: Flagship NVIDIA GPU Launch Prices (2016-2026)

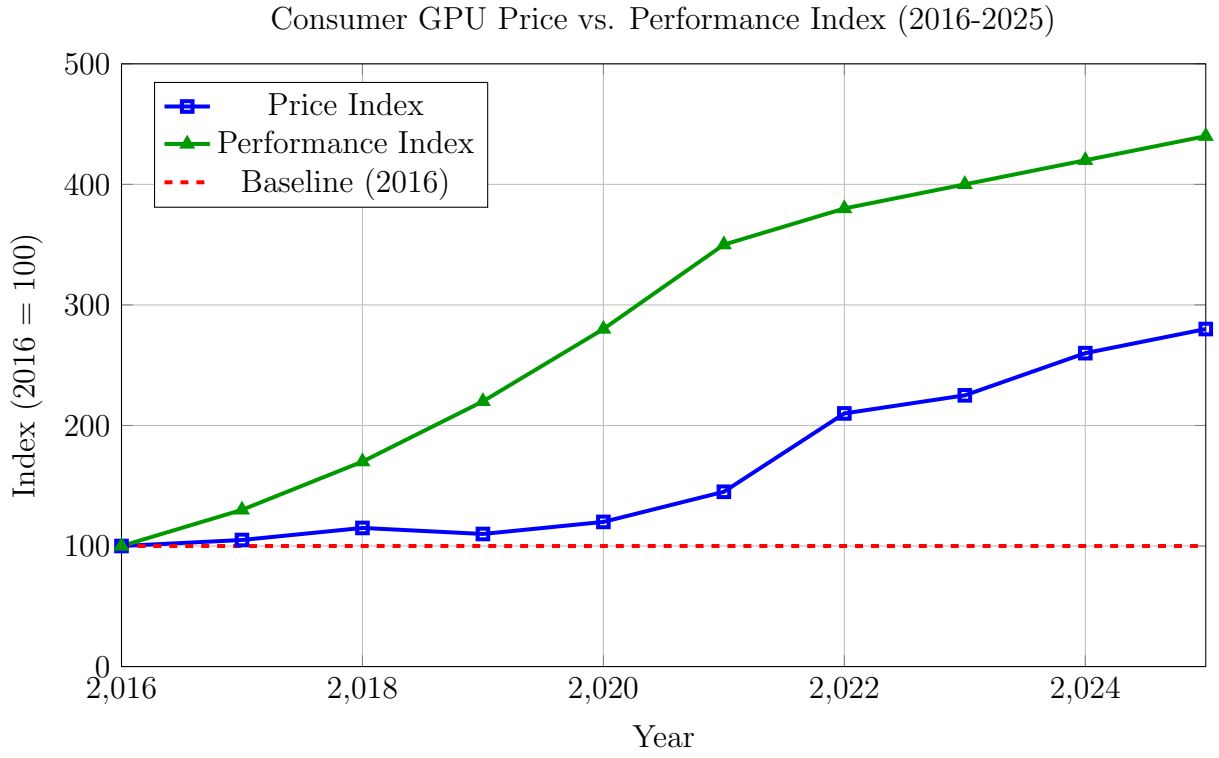


Figure 2: Consumer GPU Price vs. Performance Index (2016-2025)

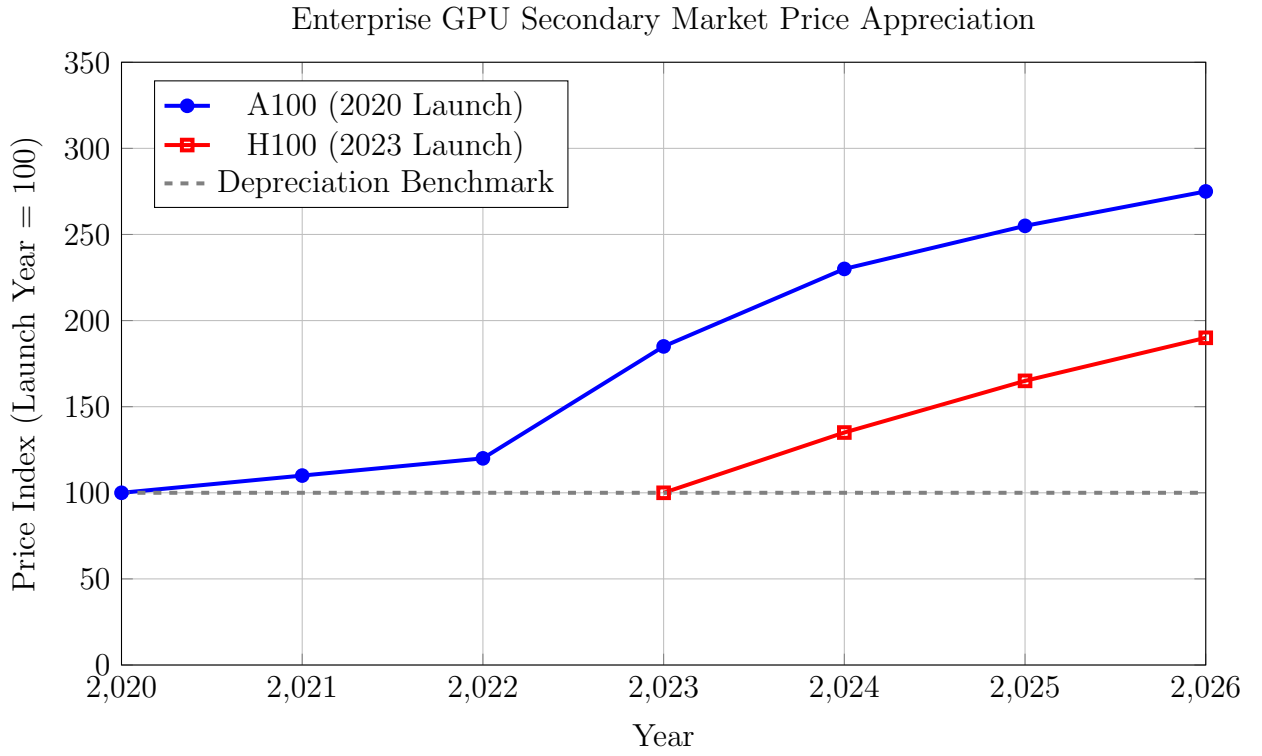


Figure 3: Enterprise GPU Secondary Market Price Index (2020-2026)

4.2 Market Composition Tables

Table 1 illustrates the shift in consumer market composition. The sub-\$300 segment has contracted significantly from 45% in 2019 to 28% in 2026, while the \$1000+ segment has expanded from 3% to 15% over the same period. This reflects both the general price inflation across all GPU tiers and the upward pressure on consumer pricing driven by enterprise demand.

Table 1: GPU Price Tiers and Market Share (Consumer Segment)

Price Range	2019 Share	2023 Share	2026 Share	Change
\$0-\$299	45%	32%	28%	-17%
\$300-\$599	40%	38%	35%	-5%
\$600-\$999	12%	20%	22%	+10%
\$1000+	3%	10%	15%	+12%

Table 2 shows the reallocation of data center capital expenditure toward storage, particularly HBM. The storage category’s share is projected to increase from 25% in 2022 to 32% in 2026, reflecting the memory bottleneck that has become a defining feature of the LLM-era GPU market, as discussed in Section 3.1.

Table 2: Corporate GPU Capex Allocation 2022 vs. 2026

Category	2022 Share	2026 Share (Projected)
Computing (GPU/CPU)	55%	48%
Storage (SSD, HBM)	25%	32%
Networking	12%	12%
Other Infrastructure	8%	8%

5 Emerging Trends and Future Directions

5.1 The Shift from Training to Inference

The AI industry is currently undergoing a transition from the *training phase* to the *inference phase*. Industry estimates suggest that 70-80% of future AI compute demand will come from inference workloads rather than model training (Zhang and Wang, 2025). This transition has important implications for the GPU market.

Inference workloads differ from training in their performance characteristics. They are less demanding in terms of raw computational throughput but more sensitive to latency, memory bandwidth, and cost per query. This creates opportunities for more diverse hardware solutions. Older enterprise GPUs, such as the V100, can be repurposed for inference tasks where their memory bandwidth characteristics make them cost-effective alternatives to newer generations (36Kr, 2025). The used enterprise GPU market has thus become an important secondary supply channel.

5.2 The Rise of Domestic Alternatives

The inference transition also creates opportunities for domestic GPU manufacturers. In 2025, Chinese domestic AI accelerator suppliers achieved a 41% market share in China’s

AI accelerator market (36Kr, 2025). The shift from training to inference favors domestic vendors because inference workloads are more tolerant of software stack incompatibility and are more sensitive to price-performance ratios than absolute peak performance.

The CUDA ecosystem has long been considered NVIDIA’s primary competitive advantage, but the inference phase reduces the criticality of this advantage. Inference workloads can be efficiently deployed on hardware from multiple vendors, provided the software stack is optimized for the specific workload (36Kr, 2025). This has opened a window for domestic alternatives to gain market share.

5.3 The Energy Dimension

One emerging constraint on GPU deployment that has not yet received sufficient attention is electricity. The concept of *intelligence per kilowatt-hour* is becoming increasingly relevant as data center operators face both regulatory pressure and cost pressure on energy consumption (China National Radio, 2026). The GPU market of the future will need to consider not just performance and price but also energy efficiency. This may drive a reorientation toward more efficient architectures and specialized inference chips.

6 Conclusion

The LLM revolution has fundamentally restructured the GPU market. The pre-LLM era, where gaming demand anchored pricing and consumer electronics followed predictable depreciation patterns as described in Section 2, has been replaced by an era where GPUs function as hard currency in the global AI arms race. We have identified three structural drivers of this transformation: exponential demand growth from enterprise AI deployment, persistent supply constraints driven by HBM memory bottlenecks, and the emerging financialization of GPUs as collateralizable assets.

The implications of this transformation are far-reaching. For consumers, the upgrade cycle is disrupted and the concept of *value for money* has shifted dramatically. The bifurcation documented in Section 3.2 shows that enterprise buyers face an entirely different market than individual consumers. For enterprises, GPU acquisition has become a strategic imperative, and the inability to secure sufficient GPU capacity creates an existential risk for AI business models. For policymakers, the GPU supply chain represents a critical infrastructure vulnerability with national security dimensions.

Looking forward, we anticipate the transition from training to inference, discussed in Section 5.1, will reshape the market once again. The inference phase is likely to be characterized by greater hardware diversity, increased price sensitivity, and the emergence of specialized inference chips. The rise of domestic GPU alternatives, covered in Section 5.2, particularly in China’s market, suggests a future with a more fragmented supply base. The energy efficiency dimension, captured in the concept of *intelligence per kilowatt-hour* (Section 5.3), will become increasingly salient as data center scale grows.

The GPU market of the future will not return to the pre-LLM status quo. The demand base has permanently expanded, the supply constraints are structural rather than transitory, and the financial value of compute capacity is now broadly understood. The question for all stakeholders is not whether the GPU market will revert to historical patterns, but rather how the new equilibrium will be established in a world where AI compute demand shows no signs of abating.

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